Note 2.3 Additional Topics and Applications

The Adjoint of a Matrix

- **Definition:**
  Let A be a $n \times n$ matrix. We define a new matrix called the adjoint of A by

$$\text{adj}A = \begin{pmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & & & \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{pmatrix}$$

- **Properties:**
  By Lemma 2.2.1,

$$A(\text{adj}A) = \det(A)I$$

If A is nonsingular, $\det(A)$ is a nonzero scalar, then

$$A\left(\frac{1}{\det(A)}\text{adj}A\right) = I$$

Thus,

$$A^{-1} = \frac{1}{\det(A)}\text{adj}A, \det(A) \neq 0$$

Cramer's Rule

- **Theorem 2.3.1 Cramer's Rule:**
  Let A be an $n \times n$ nonsingular matrix, and let $\mathbf{b} \in \mathbb{R}^n$. Let A_i be the matrix replacing the ith column of A by $\mathbf{b}$. If $\mathbf{x}$ is the unique solution of $A\mathbf{x} = \mathbf{b}$, then

$$x_i = \frac{\det(A_i)}{\det(A)} \quad for \quad i = 1, 2, \cdots, n$$

Proof:
Since

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{\det(A)}(\text{adj}A)\mathbf{b}$$

then

$$\mathbf{x} = \begin{pmatrix} b_1A_{11} + b_2A_{21} + \cdots + b_nA_{n1} \\ b_1A_{12} + b_2A_{22} + \cdots + b_nA_{n2} \\ \vdots \\ b_1A_{1n} + b_2A_{2n} + \cdots + b_nA_{nn} \end{pmatrix}$$

thus

$$\begin{aligned} x_i &= \frac{b_1A_{1i} + b_2A_{2i} + \cdots + b_nA_{ni}}{\det(A)} \\ &= \frac{\det(A_i)}{\det(A)} \end{aligned}$$

Cramer's Rule gives us a method for writing the solution of $n \times n$ system of linear equations in terms of determinants.

Application 1 Coded Messages

- **Idea:**
  Take a matrix A whose entries are all integers and whose determinant is 1.(ensure the entries in the A^{-1} be integers) Then multiply the matrix with coded message by A to further encode the coded message. Then the message can be decode by multiplying the matrix by A^{-1}. (The matrix A can be obtained by apply row operation 1 and 3 to the identity I).

The Cross Product

- **Definition:**
  Given two vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^3$, then we can define the cross product, denoted $\mathbf{x} \times \mathbf{y}$,

$$\mathbf{x} \times \mathbf{y} = \begin{pmatrix} x_2y_3 - y_2x_3 \\ y_1x_3 - x_1y_3 \\ x_1y_2 - y_1x_2 \end{pmatrix}$$

If C is any matrix of the form

$$C = \begin{pmatrix} i & x_1 & y_1 \\ j & x_2 & y_2 \\ k & x_3 & y_3 \end{pmatrix}$$

then $\det(C) = iC_{11} + jC_{21} + kC_{31} = (\mathbf{x} \times \mathbf{y})$ where $= (i, j, k)$. Then

$$\mathbf{x} \times \mathbf{y} = C_{11}\mathbf{e_1} + C_{21}\mathbf{e_2} + C_{31}\mathbf{e} = \begin{pmatrix} C_{11} \\ C_{21} \\ C_{31} \end{pmatrix} = \begin{vmatrix} \mathbf{e}_1 & x_1 & y_1 \\ \mathbf{e}_2 & x_2 & y_2 \\ \mathbf{e}_3 & x_3 & y_3 \end{vmatrix}$$

Application 2 Newtonian Mechanics

- **Definition of length of vector:**
  If $\mathbf{x}$ is a vector in either $\mathbb{R}^2$ or $\mathbb{R}^3$, then we define the length of $\mathbf{x}$, denoted $||x||$, by

  $$||\mathbf{x}|| = (\mathbf{x}^T\mathbf{x})^{12}$$

  and a vector is said to be a unit vector if $||\mathbf{x}|| = 1$.
- **The Representation of a Vector in Two-or-Three-Dimensional Space**
  - **In two-dimensional space:**
    We can conveniently represent the position of vectors at time t as linear combination of vectors (t) and (t), where (t) is a unit vector in the direction of the tangent line to curve at the point $(x_1(t), x_2(t))$ and (t) is a unit vector in the direction of a normal line to the curve at the given point.
  - **In three-dimensional space:**
    We define the third unit vector $(t) = (t) \times (t)$ as the binormal vector which obeys the right hand rule.

Exercise Section 2.3

- Question 6th
  Property:
  If A is singular, the product $A\,\mathrm{adj}A = 0$.
- Question 8th
  Property:
  If A is a nonsingular $n \times n$ matrix with $n > 1$, then

  $$\det(\mathrm{adj}\ A) = (\det(A))^{n-1}$$

- Question 10th
  Property:
  If A is nonsingular, then $\mathrm{adj}\ A$ is nonsingular and

  $$(\mathrm{adj}\ A)^{-1} = \det(A^{-1})A = \mathrm{adj}\ A^{-1}$$

- Question 11th
  Suppose $\mathrm{adj}\ A$ is nonsingular, then there exist $(\mathrm{adj}\ A)^{-1}$. Then

  $$A\,\mathrm{adj}\ A = \det(A)I = O$$

  Then multiply $\mathrm{adj}\ A$ of both sides, then

$$A = O$$

But when $A = O, \operatorname{adj} A = O$, which is singular. That is conflict with the hypothesis.